

A PROPOSED SMALL-RESOLUTION PROOF OF CONJECTURE 6.4

ABSTRACT. This note records a proposed proof of the formula

$$P_{e,w}(q) = (1 + q + \cdots + q^\ell)^{k-1}$$

for the permutations appearing in Conjecture 6.4 of *Big Data Approach to Kazhdan–Lusztig Polynomials*. The proof constructs an explicit incidence resolution and checks smallness by a dimension estimate for rank-excess strata.

1. STATEMENT AND CONVENTIONS

Let $k \geq 1$ and $\ell \geq 0$, and put

$$n = k + 2\ell, \quad m = k - 1.$$

Let $V = \mathbb{C}^n$, and let

$$E_i = \langle e_1, \dots, e_i \rangle$$

be the standard flag. We consider the permutation

$$w = [n, n-1, \dots, k+\ell+1, \ell+1, \ell+2, \dots, \ell+k, \ell, \ell-1, \dots, 1] \in S_n.$$

This is the permutation denoted w_ℓ^k in the convention where

$$w_\ell^k = (w_0^\ell \otimes \text{id}_k \otimes w_0^\ell) \circ u_\ell^k.$$

If the opposite convention for composition is used, one obtains w^{-1} instead; this does not change the final Kazhdan–Lusztig polynomial, since $P_{x,w} = P_{x^{-1},w^{-1}}$.

We use the standard geometric convention that $P_{e,w}(q)$ is the local intersection-cohomology Poincaré polynomial of the Schubert variety X_w at the standard flag $e = E_\bullet$, with q recording cohomological degree 2.

Theorem 1. *For the permutation w above,*

$$P_{e,w}(q) = (1 + q + \cdots + q^\ell)^{k-1}.$$

For $\ell = 0$, the permutation is the identity and the formula is immediate. For $k = 1$, one has $m = 0$ and $w = w_0$, so $X_w = \text{Fl}(V)$ is smooth and again the formula is immediate. We therefore assume from now on that $\ell \geq 1$ and $k \geq 2$.

2. THE SCHUBERT CONDITIONS

For p, q , define

$$r_w(p, q) = \#\{a \leq p \mid w(a) \leq q\}.$$

The Schubert variety $X_w \subset \text{Fl}(V)$ is given by

$$X_w = \{F_\bullet \mid \dim(F_p \cap E_q) \geq r_w(p, q) \text{ for all } p, q\}.$$

We first identify the essential conditions for this particular w .

Lemma 1. *The Schubert variety X_w is the closed subvariety of $\text{Fl}(V)$ defined by*

$$\dim(F_{\ell+i} \cap E_{\ell+i}) \geq i, \quad 1 \leq i \leq m = k - 1.$$

Proof. Write $a_+ = \max(a, 0)$. A direct count from the three blocks of w gives

$$r_w(p, q) = \max\{0, p + q - n, \min((p - \ell)_+, (q - \ell)_+)\}.$$

Indeed, if $p, q > \ell$ and $a = p - \ell$, $b = q - \ell$, then the contribution of the middle block together with the two reversed outer blocks is

$$\min(a, b, k) + \max(0, a - k) + \max(0, b - k) = \max\{\min(a, b), a + b - k\},$$

which is the displayed formula. The cases $p \leq \ell$ or $q \leq \ell$ reduce to the automatic term $\max(0, p + q - n)$.

Every pair of subspaces satisfies the automatic dimension bound

$$\dim(F_p \cap E_q) \geq p + q - n.$$

Thus only the term

$$t = \min((p - \ell)_+, (q - \ell)_+)$$

needs to be considered. If $1 \leq t \leq k - 1$, then $p, q \geq \ell + t$, so

$$F_{\ell+t} \cap E_{\ell+t} \subseteq F_p \cap E_q.$$

The diagonal condition for $i = t$ therefore gives

$$\dim(F_p \cap E_q) \geq t.$$

If $t \geq k$, then $p, q \geq \ell + k$. Hence

$$p + q - n = (p - \ell) + (q - \ell) - k \geq \min(p - \ell, q - \ell) = t,$$

so the automatic bound already gives the required inequality. This proves that the diagonal conditions imply all Schubert rank inequalities. The converse is immediate by taking $p = q = \ell + i$. $\square$

3. REDUCTION TO A PARTIAL FLAG VARIETY

Let

$$U_i = E_{\ell+i}, \quad 1 \leq i \leq m.$$

Let Y be the partial-flag Schubert variety

$$Y = \{H_1 \subset H_2 \subset \cdots \subset H_m \subset V \mid \dim H_i = \ell+i, \dim(H_i \cap U_i) \geq i \text{ for all } i\}.$$

The forgetful map

$$\rho : X_w \longrightarrow Y, \quad F_\bullet \longmapsto (F_{\ell+1}, \dots, F_{\ell+m})$$

identifies X_w with the inverse image of Y under the smooth projection

$$\mathrm{Fl}(V) \longrightarrow \mathrm{Fl}_{\ell+1, \dots, \ell+m}(V).$$

Thus it is enough to construct and check a small resolution of Y ; pulling back along this smooth map preserves the relevant fiber dimensions and codimensions.

4. THE INCIDENCE RESOLUTION

Define

$$\tilde{Y} = \{(H_\bullet, G_\bullet) \mid H_\bullet \in Y, 0 \subset G_1 \subset \cdots \subset G_m, \dim G_i = i, G_i \subset H_i \cap U_i \text{ for all } i\}.$$

Let

$$p : \tilde{Y} \longrightarrow Y, \quad (H_\bullet, G_\bullet) \longmapsto H_\bullet.$$

Lemma 2. *The variety $\tilde{Y}$ is smooth.*

Proof. First choose the chain $G_1 \subset \cdots \subset G_m$ with $G_i \subset U_i$ and $\dim G_i = i$. Suppose G_{i-1} has been chosen. Then G_i/G_{i-1} is a line in

$$U_i/G_{i-1},$$

which has dimension

$$(\ell + i) - (i - 1) = \ell + 1.$$

Thus each step contributes a projective space $\mathbb{P}^\ell$. The space of possible $G_\bullet$ is therefore an iterated $\mathbb{P}^\ell$ -bundle and is smooth.

Now fix $G_\bullet$. To choose $H_\bullet$, first choose H_m containing G_m and having dimension $\ell + m$. Equivalently, choose an ℓ -plane in V/G_m , which has dimension $2\ell + 1$. These choices form the Grassmannian $\mathrm{Gr}(\ell, 2\ell + 1)$.

Then choose $H_{m-1}, H_{m-2}, \dots, H_1$ descending. Once H_{i+1} has been chosen, the choice of H_i is the choice of an $(\ell+i)$ -dimensional subspace satisfying

$$G_i \subset H_i \subset H_{i+1}.$$

Equivalently, H_i/G_i is an ℓ -plane in H_{i+1}/G_i , and this quotient has dimension $\ell + 1$. Thus each such step gives $\mathrm{Gr}(\ell, \ell + 1) \cong \mathbb{P}^\ell$. Hence $\tilde{Y}$ is obtained by an iterated Grassmannian/projective-bundle construction, so it is smooth. $\square$

Lemma 3. *The map $p : \tilde{Y} \rightarrow Y$ is proper and birational.*

Proof. Properness follows because $\tilde{Y}$ is closed in a product of projective varieties over Y .

The map is surjective. If $H_\bullet \in Y$, put

$$A_i = H_i \cap U_i.$$

Then

$$A_1 \subset A_2 \subset \cdots \subset A_m, \quad \dim A_i \geq i.$$

Choose $G_1 \subset A_1$ of dimension 1, then extend inductively to an i -plane $G_i \subset A_i$ containing G_{i-1} . This is possible because A_i contains A_{i-1} and has dimension at least i .

On the open locus

$$Y^\circ = \{H_\bullet \in Y \mid \dim(H_i \cap U_i) = i \text{ for all } i\},$$

all G_i are forced:

$$G_i = H_i \cap U_i.$$

Thus p is an isomorphism over Y° . The locus Y° is nonempty: choose an ℓ -plane $B \subset V$ disjoint from U_m , and set

$$H_i = \langle e_1, \dots, e_i \rangle \oplus B.$$

Then $H_i \cap U_i = \langle e_1, \dots, e_i \rangle$. Hence p is birational. $\square$

5. FIBER DIMENSIONS

For $H_\bullet \in Y$, define the excess numbers

$$a_i = \dim(H_i \cap U_i) - i, \quad 1 \leq i \leq m.$$

Put

$$A = \sum_{i=1}^m a_i.$$

Lemma 4. *For $H_\bullet \in Y$ with excesses a_i , one has*

$$\dim p^{-1}(H_\bullet) = \sum_{i=1}^m a_i.$$

More precisely, the fiber is an iterated bundle with fibers

$$\mathbb{P}^{a_1}, \mathbb{P}^{a_2}, \dots, \mathbb{P}^{a_m}.$$

Proof. Set $A_i = H_i \cap U_i$, so $\dim A_i = i + a_i$. The fiber over $H_\bullet$ consists of chains

$$0 \subset G_1 \subset \cdots \subset G_m, \quad \dim G_i = i, \quad G_i \subset A_i.$$

Once G_{i-1} has been chosen, the quotient G_i/G_{i-1} is a line in

$$A_i/G_{i-1},$$

which has dimension

$$(i + a_i) - (i - 1) = a_i + 1.$$

Thus the i th step contributes $\mathbb{P}^{a_i}$. $\square$

6. CODIMENSION OF THE EXCESS STRATA

For a vector $a = (a_1, \dots, a_m)$ with $a_i \geq 0$, let

$$Y_a = \{H_\bullet \in Y \mid \dim(H_i \cap U_i) = i + a_i \text{ for all } i\}.$$

Some of these strata may be empty. We estimate the dimension of the nonempty ones.

Lemma 5. *For every nonempty stratum Y_a ,*

$$\text{codim}_Y(Y_a) \geq \sum_{i=1}^m (a_i^2 + 2a_i) - \sum_{i=1}^{m-1} a_i a_{i+1}.$$

Equivalently,

$$\text{codim}_Y(Y_a) \geq 2 \sum_{i=1}^m a_i + \frac{1}{2} \left(a_1^2 + a_m^2 + \sum_{i=1}^{m-1} (a_i - a_{i+1})^2 \right).$$

Proof. For $H_\bullet \in Y_a$, put

$$A_i = H_i \cap U_i, \quad \dim A_i = i + a_i.$$

Then

$$A_1 \subset A_2 \subset \dots \subset A_m, \quad A_i \subset U_i.$$

Set $a_0 = 0$. After A_{i-1} has been chosen, the choice of A_i is bounded by the choice of a subspace of dimension

$$\dim A_i - \dim A_{i-1} = 1 + a_i - a_{i-1}$$

inside

$$U_i/A_{i-1},$$

which has dimension

$$(\ell + i) - ((i - 1) + a_{i-1}) = \ell + 1 - a_{i-1}.$$

Thus the choices of the chain $A_\bullet$ contribute at most

$$\sum_{i=1}^m (1 + a_i - a_{i-1})(\ell - a_i)$$

parameters.

Now fix $A_\bullet$. First choose H_m . We require

$$A_m \subset H_m, \quad \dim H_m = \ell + m, \quad H_m \cap U_m = A_m.$$

Put

$$d_m = (\ell + m) - (m + a_m) = \ell - a_m.$$

Then H_m/A_m is a d_m -plane in V/A_m transverse to U_m/A_m . Equivalently, one chooses a d_m -plane in V/U_m , and then a linear map from that plane to U_m/A_m . Since

$$\dim(V/U_m) = \ell + 1, \quad \dim(U_m/A_m) = d_m,$$

this contributes

$$d_m((\ell + 1) - d_m) + d_m^2 = d_m(\ell + 1) = (\ell - a_m)(\ell + 1)$$

parameters.

Next choose $H_{m-1}, \dots, H_1$ descending. Once H_{i+1} has been chosen, H_i is a hyperplane in H_{i+1} containing A_i . The space of such hyperplanes has dimension

$$\dim H_{i+1} - \dim A_i - 1 = (\ell + i + 1) - (i + a_i) - 1 = \ell - a_i.$$

The additional condition $H_i \cap U_i = A_i$ is open inside this space, so it cannot increase the dimension. Therefore the choices of the lower H_i contribute at most

$$\sum_{i=1}^{m-1} (\ell - a_i)$$

parameters.

Hence

$$\dim Y_a \leq D(a),$$

where

$$D(a) = \sum_{i=1}^m (1 + a_i - a_{i-1})(\ell - a_i) + (\ell - a_m)(\ell + 1) + \sum_{i=1}^{m-1} (\ell - a_i).$$

For $a = 0$, this gives

$$D(0) = m\ell + \ell(\ell + 1) + (m - 1)\ell.$$

This is the dimension of $\tilde{Y}$, and since p is birational, it is also the dimension of Y . Thus

$$\text{codim}_Y(Y_a) \geq D(0) - D(a).$$

A direct expansion gives

$$D(0) - D(a) = \sum_{i=1}^m (a_i^2 + 2a_i) - \sum_{i=1}^{m-1} a_i a_{i+1}.$$

Finally,

$$\sum_{i=1}^m a_i^2 - \sum_{i=1}^{m-1} a_i a_{i+1} = \frac{1}{2} \left(a_1^2 + a_m^2 + \sum_{i=1}^{m-1} (a_i - a_{i+1})^2 \right),$$

which gives the equivalent form. □

Lemma 6. *The map $p : \tilde{Y} \rightarrow Y$ is small.*

Proof. Let Y_a be a non-open excess stratum, so $a \neq 0$. By the preceding two lemmas, the fiber dimension over Y_a is

$$\sum_{i=1}^m a_i,$$

while

$$\mathrm{codim}_Y(Y_a) \geq 2 \sum_{i=1}^m a_i + \frac{1}{2} \left(a_1^2 + a_m^2 + \sum_{i=1}^{m-1} (a_i - a_{i+1})^2 \right).$$

The second term is strictly positive for $a \neq 0$. Therefore

$$\mathrm{codim}_Y(Y_a) > 2 \dim p^{-1}(H_\bullet)$$

for every $H_\bullet \in Y_a$ with $a \neq 0$.

Equivalently, for every $d > 0$, the closed locus

$$\{H_\bullet \in Y \mid \dim p^{-1}(H_\bullet) \geq d\}$$

is a finite union of such strata with $\sum_i a_i \geq d$, and each has codimension strictly larger than $2d$. This is precisely the smallness condition. $\square$

7. PULLBACK TO COMPLETE FLAGS

Let

$$\tilde{X} = X_w \times_Y \tilde{Y}.$$

Since $X_w \rightarrow Y$ is the pullback of the smooth projection from the complete flag variety to the relevant partial flag variety, the map

$$\pi : \tilde{X} \rightarrow X_w$$

is the smooth base change of $p : \tilde{Y} \rightarrow Y$. Hence $\tilde{X}$ is smooth, π is proper and birational, and smallness is preserved under this pullback: the fibers are the same as those of p , and the codimensions of the corresponding strata are unchanged.

Thus $\pi : \tilde{X} \rightarrow X_w$ is a small resolution of singularities.

8. THE FIBER OVER THE STANDARD FLAG

At the standard flag $e = E_\bullet$, one has

$$F_{\ell+i} = E_{\ell+i} = U_i.$$

Therefore

$$\pi^{-1}(e) = \{0 \subset G_1 \subset \cdots \subset G_m \mid \dim G_i = i, G_i \subset E_{\ell+i}\}.$$

After G_{i-1} has been chosen, the quotient G_i/G_{i-1} is a line in

$$E_{\ell+i}/G_{i-1},$$

which has dimension

$$(\ell + i) - (i - 1) = \ell + 1.$$

Thus $\pi^{-1}(e)$ is an iterated $\mathbb{P}^\ell$ -bundle of length $m = k - 1$. By the projective bundle formula,

$$\sum_j \dim H^{2j}(\pi^{-1}(e), \mathbb{C}) q^j = (1 + q + \cdots + q^\ell)^m.$$

Because π is small,

$$\mathrm{IC}_{X_w} \cong R\pi_* \mathbb{C}_{\tilde{X}}[\dim X_w].$$

Hence the local intersection cohomology of X_w at e is the ordinary cohomology of the fiber $\pi^{-1}(e)$. With the standard geometric normalization of Kazhdan–Lusztig polynomials, this gives

$$P_{e,w}(q) = (1 + q + \cdots + q^\ell)^m = (1 + q + \cdots + q^\ell)^{k-1}.$$

This proves the theorem.

Remark 1. *The argument above is not the naive replacement of the $\mathbb{P}^1$ -fibers in the Shapiro–Shapiro–Vainshtein resolution by $\mathbb{P}^\ell$ -fibers in the same two-step geometry. The incidence variety here uses the whole auxiliary chain*

$$G_1 \subset \cdots \subset G_{k-1}, \quad \dim G_i = i,$$

which is what makes the smallness estimate possible.

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